

(1) Write a Matlab or Python program that can convert a numbered musical notation (簡譜) into a music file (*.wav).

Example: (Twinkle twinkle little stars)

```
score = [1, 1, 5, 5, 6, 6, 5]; % 1: Do, 2: Re, 3: Mi, .....
```

```
beat=[ 1, 1, 1, 1, 1, 1, 2]; % 拍子
```

```
name= 'twinkle';
```

```
getmusic(score, beat, name) % generate the music file twinkle.wav
```

The Matlab / Python code should be handed out by NTUCool.

With basic requirement (score, beat, name): 24 scores

程式的功能越多，考慮的因素越多，分數越高

程式的功能要清楚說明以方便助教批改

(30 scores)

功能：

1. 出現兩個連續音，第2個音振幅增加
2. 開頭漸強、結尾漸弱
3. 倍頻
4. 音量調整
5. 斷音
6. 重播一次升高兩個半音，換D大調。

(2) (a) Determine $2^{700} \bmod 67$.

(b) Find an integer x between 0 and 2800 that satisfies (i) $x \bmod 43 = 4$ and (ii) $x \bmod 67 = 15$
(Hint: Using the Chinese remainder theorem).

(c) Determine $39! \bmod 43$ (Hint: Using the Wilson theorem). (12 scores)

(a) $2^{700} \bmod 67$

$$(2^{66})^{10} \times 2^{40} \bmod 67 = 1 \times 2^{40} \bmod 67$$

40 bit binary 101000 $\rightarrow 2^5 + 2^3$
 $\rightarrow 2^{40} = 2^{32} \cdot 2^8$

$$2^8 \bmod 67 = 256 \bmod 67 = 55$$

$$2^{32} \bmod 67 = (2^8)^4 \bmod 67 = 33$$

$$2^{40} \bmod 67 = 55 \times 33 \bmod 67 = 6$$

$$2^{700} \bmod 67 = 6 *$$

(b) $a = 43 \nmid \bmod 67 = 15$

$$53 \times 43 \bmod 67 = 1 \rightarrow 53 \times 43 \times 15$$

$$b = 67 \nmid \bmod 43 = 9$$

$$9 \times 67 \bmod 43 = 1 \rightarrow 9 \times 67 \times 9$$

$$53 \times 43 \times 15 + 9 \times 67 \times 9 \bmod 67 \times 43$$

$$2412 + 34185 \bmod 2881$$

$$36597 \bmod 2881$$

$$36597 \div 2881 = 12 \dots$$

$$36597 - 2881 \times 12 = 2025 *$$

• 43 is mod 67 bit inverse = 53

$$67 = 1 \times 43 + 24$$

$$5 - 1 \times 4 = 1$$

$$43 = 1 \times 24 + 19$$

$$5 - (19 - 3 \times 5) = 1$$

$$24 = 1 \times 19 + 5$$

$$5 \times 4 - 1 \times 19 = 1$$

$$19 = 3 \times 5 + 4$$

$$(24 - 19 \times 1) \times 4 - 1 \times 19 = 1$$

$$5 = 1 \times 4 + 1$$

$$24 \times 4 - 5 \times 19 = 1$$

$$24 \times 4 - 5 \times (43 - 1 \times 24) = 1$$

$$24 \times 9 - 5 \times 43 = 1$$

$$9 \times (67 - 1 \times 43) - 5 \times 43 = 1$$

$$9 \times 67 - 14 \times 43 = 1$$

$$14 \times 43 \bmod 67 = 1$$

$$67 - 14 = 53$$

• 67 is mod 43 bit inverse = 9

$$67 \bmod 43 = 24$$

$$24 \nmid \bmod 43 = 1$$

$$24 \times 9 \bmod 43 = 1$$

(c) $(p-1)! \bmod p = -1$ $42! \bmod 43 = -1$

$$39! \times 40 \times 41 \times 42 \bmod 43 = -1$$

40, 41, 42 $\rightarrow 39! \times 40 \times 41 \times (-1) \bmod 43 = -1640 \bmod 43$ $1640 \div 43 = 38 \dots$

mod 43

$$1640 - 43 \times 38 = 6 \Rightarrow 40 \times 41 \times 42 \bmod 43 = -6$$

$$39! \cdot (-6) \bmod 43 = -1$$

$$\Rightarrow 39! \bmod 43 = 6^{-1}$$

$$6 \nmid \bmod 43 = 1, \Rightarrow 6 \times 36 \bmod 43 = 1$$

$$39! \bmod 43 = 36 *$$

(3) Given $M = 11$, $\alpha = 8+6i$, and $N = 12$. Determine the complex number theoretic transform (CNT) of x where

$$x = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \end{matrix} \\ \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 4 & 0 & 2 & 0 & 2 & 0 & 4 & 0 & 2 & 0 & 2 & 0 \end{bmatrix}^T \end{matrix}$$

Hint: $\text{fft}(x)$ is as follows. It is Ramanujan's sum.

$$\text{fft } x = \begin{bmatrix} 4 & 0 & 2 & 0 & -2 & 0 & -4 & 0 & -2 & 0 & 2 & 0 \end{bmatrix}^T \quad (8 \text{ scores})$$

$$\text{fft}(x) = \begin{bmatrix} 4 & 0 & 2 & 0 & -2 & 0 & -4 & 0 & -2 & 0 & 2 & 0 \end{bmatrix}^T \quad -2 \text{ Mod } 11 \Rightarrow 9$$

$$\text{fft}(x) = \begin{bmatrix} 4 & 0 & 2 & 0 & 9 & 0 & 7 & 0 & 9 & 0 & 2 & 0 \end{bmatrix}^T \quad -4 \text{ Mod } 11 \Rightarrow 7$$

(4) What is the Legendre sequence corresponding to $p = 11$? (Hint: The sequence should have 11 entries). (10 scores)

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{如果 } a \bmod p = 0 \\ 1 & \text{如果 } a \text{ 是 } \bmod p \text{ 二次剩餘} \\ -1 & \text{如果 } a \text{ 不是 } \bmod p \text{ 二次剩餘} \end{cases}, \quad a_n = \left(\frac{n}{p}\right), \quad n = 0, 1, 2, \dots, p-1$$

$$a = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$$

$$a^2 \bmod 11 = 0, 1, 4, 9, 5, 3, 3, 5, 9, 4, 1$$

$$\text{平方剩餘: } 0, 1, 3, 4, 5, 9,$$

$$\text{legendre sequence: } [0, 1, -1, 1, 1, -1, -1, -1, -1, 1, -1]$$

(5) Suppose that there is a multipath system $y[n] = x[n] + 0.4x[n-20] + 0.2x[n-30]$.

(a) Find $p[n]$ such that $y[n] = x[n] * p[n]$.

(b) Design the lifter to remove the effect of $p[n]$ and try to not destroy $x[n]$ as possible. (10 scores)

$$(A) \quad y[n] = x[n] + 0.4x[n-20] + 0.2x[n-30].$$

$$p[n] = \delta[n] + 0.4\delta[n-20] + 0.2\delta[n-30].$$

$$P(Z) = 1 + 0.4Z^{-20} + 0.2Z^{-30}$$

$$\hat{P}(Z) = \log(1 + 0.4Z^{-20} + 0.2Z^{-30}) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{(0.4Z^{-20} + 0.2Z^{-30})^k}{k}$$

$$\approx (0.4Z^{-20} + 0.2Z^{-30}) - \frac{1}{2}(0.16Z^{-40} + 0.16Z^{-50} + 0.04Z^{-60}) \dots$$

$$\hat{p}[n] = 0.4\delta[n-20] + 0.2\delta[n-30] - 0.08\delta[n-40] - 0.08\delta[n-50] - 0.02\delta[n-60] \dots *$$

$$(b) \quad y[n] = x[n] * h[n]$$

$$h[n] = \delta[n] - 0.4\delta[n-20] - 0.2\delta[n-30] *$$

(6) (a) Is it possible for humans to hear the voice with the frequency of 19Hz? Why?

(b) In the noiseless case, in what condition we cannot use the variation of amplitude to separate a speech signal into several syllables?

(c) Why a music signal always has the chord (和弦) phenomenon? (15 scores)

(a) 人耳的感知能力就像 bandpass filter, pass band 是落在 20 ~ 20000 Hz, 19 Hz 基本上是聽不到

(b) 1. 在沒有子音的時候, 這個音不是由子音能量小, 振幅小作開頭, 會和前一個音的母音幅幅相近.

2. 用漢語拼音或羅馬拼音長 雙母音 和 母音 + 濁音, 也會有相同狀況.

(c) 因為倍頻現象.

$$\text{和弦長 } L \rightarrow L = \frac{n}{2}\lambda, \quad \lambda = \frac{2L}{n}$$

$$f = \frac{n}{2} \cdot \frac{340}{L} = nf_0, \quad f_0 = \frac{340}{2L}$$

(7) (a) What is the way to measure the uniformity (一致性) in mathematics?

(b) Why the compression ratio of an image can be higher than that of the vocal signal?

(c) In addition to the DCT, which is adopted by MP3, write at least three possible ways that can compress a music signal more efficiently. (15 scores)

(a) 可用 $entropy = -\sum_{i=1}^n P_i \log P_i$, $P_1, P_2, P_3, \dots, P_n$ the probability for each case.

(b) Audio $\rightarrow$ 1 Dimension 可以用來壓縮的資訊較少, 每個點相關性不高
難預設下一個點的值, 一致性低.

image $\rightarrow$ 2 Dimension 可以用來壓縮的資訊會比 1 Dimension 的多,
和在空間上有相對高的一致性.

(c) 1. Wavelet Transform

2. Perceptual Coding (Psychoacoustic Models)

3. Linear Predictive Coding

• Extra Q: 音樂訊號和語音訊號, 哪一種比較容易辨識?

A: 音樂訊號較容易辨識.

因為音樂的音排列組合比語音少